

JINGYUAN LIU

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RESEARCH INTERESTS

Vision-Language-Action Models, Reward Modeling and Reinforcement Learning for Robotics, World Models for Robotic Manipulation, Continual and Self-Improving Robot Learning

EDUCATION

Fudan University, Shanghai, China Sep. 2023 – Jun. 2027 (Expected)
B.S. in Software Engineering [Transcript] GPA: 3.74/4.00 Rank: 28/109

University of Toronto, Toronto, Canada Sep. 2025 – Dec. 2025
Exchange Student, Faculty of Applied Science & Engineering [Transcript] GPA: 4.00/4.00

PUBLICATIONS & MANUSCRIPTS

- **ACE-Brain Team.** “ACE-Brain-0.5: A Unified Embodied Foundation Model for Physical Agentic AI.” *arXiv preprint arXiv:2607.04426*, 2026.
- **Jingyuan Liu**, et al. “Runtime Monitoring Framework for Failure Prediction and Detection.” *Manuscript in preparation for ICLR 2027*, 2026.
- **Jingyuan Liu**, et al. “Bridging World Model and Inverse Dynamics Model with Wrist Camera.” *Manuscript in preparation*, 2026.

RESEARCH EXPERIENCE

ACE Robotics, SenseTime, Shanghai, China Jan. 2026 – Present
Robotics Algorithm Intern Advisor: Zhi Hou

- **Reward Modeling for Robotic Manipulation.** Built large-scale training datasets and evaluation benchmarks for progress estimation and reward modeling, and trained specialized models integrated into **ACE-Brain-0.5**.
- **Runtime Monitoring for Embodied Agents.** Leading first-author research on a policy-decoupled runtime monitoring framework for robotic manipulation, addressing both proactive anticipation of execution failures and rapid identification of failures after onset.
- **Wrist-Camera-Centered Manipulation.** Leading first-author research on a wrist-camera-centered manipulation paradigm that uses wrist-view observations as an interface between world models and 3D pose estimation, leveraging predicted observation priors for action estimation.
- **Policy Evaluation in Simulation and the Real World.** Built evaluation pipelines for embodied policies across LIBERO, RoboCasa365, and RoboLab. Conducted real-world rollout, evaluation, and debugging on AgileX and Galaxea GR1 Pro embodiments.

STARS Laboratory, University of Toronto, Toronto, Canada Nov. 2025 – Jan. 2026
Student Research Intern Advisor: Jonathan Kelly

- Built and validated a Rubik’s Cube perception dataset for pose and face-color estimation in robotic manipulation.

Fudan Vision and Learning Laboratory, Shanghai, China Feb. 2025 – Sep. 2025
Student Research Intern Advisor: Zuxuan Wu

- Explored optical flow as affordance representations for visual perception and decision-making in robotic manipulation.

- Investigated the generalization of diffusion-based imitation policies under controlled dataset shifts using trajectories generated with MimicGen.
- Added language-conditioned task inputs to 3D Diffusion Policy and evaluated its performance in multi-task robotic manipulation settings.

SELECTED COURSE PROJECTS

- **Robot Navigation System:** Built a ROS-based navigation stack integrating Dijkstra and A* for global path planning with VFH for reactive obstacle avoidance and local navigation.
- **World Model Reproduction and Extension:** Reproduced the World Models framework and modernized its perception and dynamics modules with VQ-VAE and a causal Transformer; systematically evaluated VAE/VQ-VAE and RNN/Transformer variants on CarRacing and Breakout through reconstruction, latent prediction, and environment rollouts. [Code]
- **LLM Agent for Inference Runtime Optimization:** Built an autonomous agent to iteratively generate and optimize LLM inference runtimes, using correctness testing and performance feedback to guide code refinement and GPU operator optimization.

ADDITIONAL SKILLS

- **English:** IELTS 7.5; TOEFL 101; Duolingo English Test 145
- **Programming:** Python, C++, C, Java
- **Deep Learning & Robotics:** Hands-on experience in model reproduction, simulation-based policy evaluation, and real-world robot deployment and debugging.

HONORS & AWARDS

Fudan University Outstanding Student Award	2024, 2025
Mathematical Contest in Modeling (MCM), Honorable Mention	2025